

From Language Priors towards Physical AGI

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A model may explain how to open a drawer yet fail when its handle, friction, or contents change. This gap between knowing what usually works and discovering what works here motivates the transition from language priors to physical agency. Physical AI concerns systems that perceive, predict, reason about, or act in physical environments; Physical AGI raises the stronger, still aspirational question of whether such competence can generalize across unfamiliar tasks, environments, dynamics, and embodiments with limited new experience. We develop an analytical framework for this transition around five functional interfaces: world knowledge, multimodal state grounding, action abstraction, predictive world modeling, and integrated embodied agency. Language priors supply semantic and procedural structure, but reliable interaction also requires action-relevant state, feedback, and mechanisms for learning from discrepancies between expected and observed outcomes. These functions need not reside in separate models, and assembling them does not by itself establish general intelligence. We therefore separate architectural coverage from empirical evidence of generality, specifying how held-out task families, bounded adaptation, skill retention, and cross-embodiment transfer can make progress testable. Published benchmark coverage statistics illustrate the available evaluation substrate without conflating task counts with agent capability. We also distinguish frontier digital-agent capabilities from demonstrated physical competence. The resulting research agenda connects grounding and prediction to transferable learning, persistent memory, reliable recovery, and reproducible closed-loop evaluation. Our central argument is that progress towards Physical AGI should be measured by what an agent can learn and reliably accomplish beyond its training conditions, not by model scale or the number of components in its architecture.

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Resources: <https://github.com/Hai-chao-Zhang/Awesome-Physical-AI>

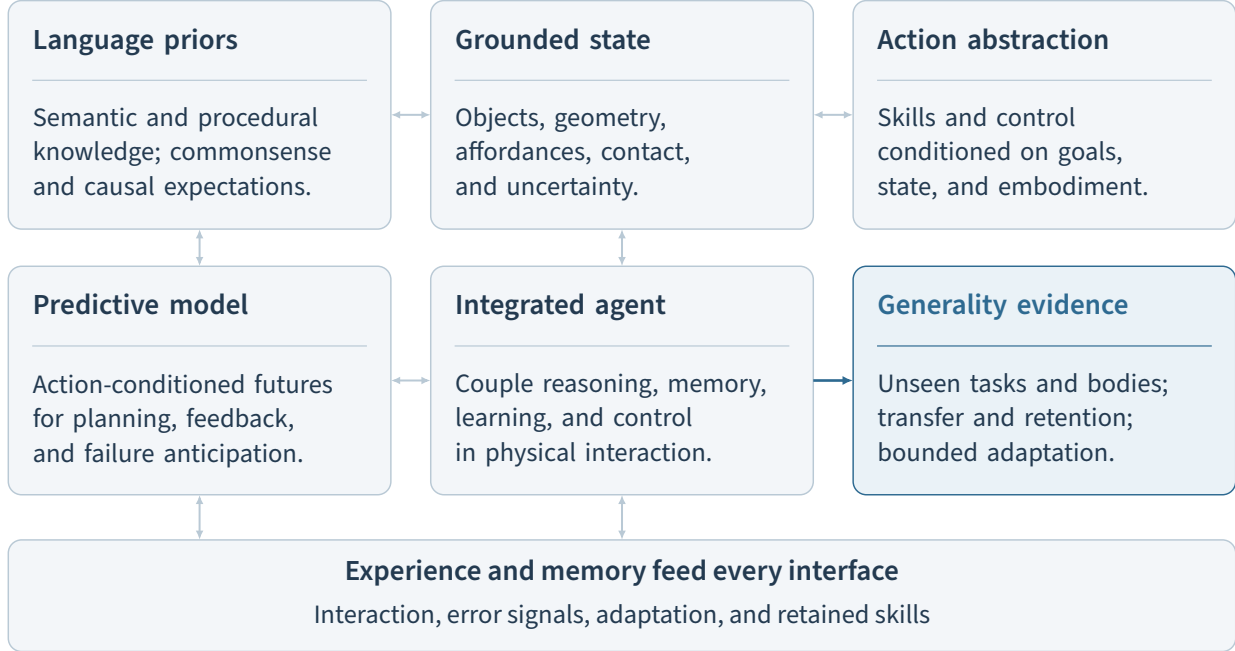
1 Introduction

Consider an agent asked to put a fragile cup into an unfamiliar drawer. A language model can suggest a sensible sequence: locate the cup, open the drawer, grasp gently, and place the cup inside. Yet a correct description does not determine whether the handle is reachable, how much force is needed, or whether the cup will collide with an unseen object. If the drawer jams, the agent must recognize that its expectation was wrong, gather new evidence, and change its behavior. The distance between this plausible plan and reliable physical execution is the starting point of this paper.

Large language models (LLMs), including GPT-4 (Achiam et al., 2023), Gemini (Team et al., 2023), and Claude (Anthropic, 2024), provide a powerful source of task-level structure. Instruction following and reasoning over intermediate steps make language useful for expressing goals and decomposing problems (Brown et al., 2020; Wei et al., 2022). Pretraining also yields parametric knowledge about objects, relations, and recurring situations (Petrone et al., 2019; Roberts et al., 2020; Kandpal et al., 2023; Chang et al., 2024). We call the semantic, commonsense, procedural, and causal expectations that an agent draws from such models *language priors*. A prior suggests what may be useful; it is not a measurement of the present scene or a guarantee about an action’s consequences.

From language priors toward physical generality

Functional interfaces may be coupled, revisited, or implemented jointly.



Assess autonomy, recovery, and safety separately from generality.

Figure 1 From language priors towards Physical AGI. Five functional interfaces connect prior knowledge with grounded state, action, prediction, and integrated agency. They may be implemented jointly and need not form a serial pipeline. The sixth panel is an evidence requirement, not another architectural component: generality must be tested under novelty and bounded adaptation. Experience and memory can update multiple interfaces, while autonomy and safety require separate assessment.

From descriptions to consequences. Physical interaction exposes a representational mismatch. Natural-language descriptions often omit metric geometry, contact, velocity, uncertainty, and fast changes that matter to control. This is a limitation of the chosen interface, not a proof that a language-based model can never represent physical variables: programs, structured states, and tool calls can carry information that prose leaves implicit. The central question is how semantic priors should be grounded, tested, and revised when decisions have physical consequences.

Vision-language models (VLMs) and multimodal LLMs connect those priors to observations (Alayrac et al., 2022; Li et al., 2023b; Liu et al., 2023c; Team et al., 2023; Lu et al., 2025). Recognizing a handle, however, is different from estimating the configuration in which pulling it succeeds. Work on intermediate visual features and dense physical perception highlights the importance of representations beyond a final textual answer (Zhang et al., 2025b,a). Vision-language-action (VLA) models make the next interface explicit by connecting observations and instructions to executable behavior (Driess et al., 2023; Zitkovich et al., 2023; Kim et al., 2025; Black et al., 2025). Their success raises a further question: what happens when the next state falls outside the situations represented by their training experience?

Predictive world models offer one mechanism for reasoning about such consequences (Ha and Schmidhuber, 2018; LeCun, 2022). They can support action-conditioned prediction, planning, and the diagnosis of unexpected outcomes. Explicit predictive modules are not mandatory for every competent policy, and world modeling need not be separate from action generation. The useful distinction is functional: proposing an action, predicting its effects, and verifying what actually happened are different obligations even when one network performs all three.

From physical competence to physical generality. *Physical AI* names a domain of capability: perceiving, predicting, reasoning about, or acting in physical environments. *Physical AGI* names a stronger research objective: broad cognitive and physical competence that transfers to unfamiliar tasks and conditions, acquires new skills efficiently, and retains useful knowledge across experience. A robot can be excellent at one distribution of tasks without meeting that objective. Conversely, a more capable digital assistant does not establish physical generality unless its capabilities survive grounding, embodied constraints, and closed-loop evaluation. We use *towards* deliberately: the framework organizes evidence and open problems, rather than certifying any existing system as Physical AGI.

Scope and contributions. We analyze language priors as one important route to physical generality, not as a necessary foundation for all robotics or intelligence. The discussion connects language-based knowledge, multimodal grounding, action interfaces, predictive models, and embodied deployment; it does not attempt exhaustive coverage of hardware, classical control, or all physical sciences. Our contribution is an analytical framework grounded in published work, not a new trained model or an experimental demonstration. It has three parts:

1. **A functional account of the grounding gap.** We distinguish five interfaces through which prior knowledge becomes physically useful, and identify what information and feedback each must preserve.
2. **A separation of system composition from generality.** We treat breadth, transfer, adaptation, and retention as empirical properties rather than consequences of a particular model family or architecture.
3. **An evidence-oriented research agenda.** We connect these properties to held-out evaluation, explicit adaptation budgets, published benchmark coverage, and failure-aware deployment, clarifying what present evidence can and cannot establish.

Figure 1 summarizes the argument. We first make the target operational, then follow the interfaces from language priors to integrated agents, and finally examine the evidence needed to support broader claims.

2 What Would Count as Progress towards Physical AGI?

The term artificial general intelligence (AGI) is useful only if it sharpens the question being asked. Work on operationalizing AGI distinguishes performance, breadth, and autonomy (Morris et al., 2023); work on measuring intelligence emphasizes skill acquisition relative to priors and experience rather than acquired skill alone (Chollet, 2019). We adapt these distinctions to physical interaction. The criteria below are a proposed analytical lens, not a consensus definition or a validated certification procedure.

A working target, not a model label. We use *Physical AGI* to denote the aspirational ability to achieve broad competence in cognitive and physical tasks, transfer knowledge across unfamiliar situations, and acquire and retain new skills under explicit resource constraints. Grounded reasoning, manipulation, navigation, and physical problem solving are relevant manifestations; a particular robot shape or neural architecture is not part of the definition. A strong claim must specify its domain and reference level of competence. Cross-embodiment transfer is evidence of breadth, but does not require every model to control every possible body without an interface or calibration.

Two axes that should not be conflated. The first axis is *functional coverage*: what the system represents, predicts, and controls. The second is *generality evidence*: how performance changes when tasks, environments, dynamics, or embodiments depart from prior experience. Adding a world model improves functional coverage only when the system actually uses its predictions; it does not automatically improve transfer. Similarly, executing more actions without help measures autonomy within a setting, not breadth across settings. Safety concerns the consequences and constraints of behavior and must be assessed alongside, rather than equated with, capability.

Novelty must have a declared meaning. For a reproducible claim, let the evaluation family be described by

$$\mathcal{F} = (\mathcal{T}, \mathcal{O}, \mathcal{E}, \mathcal{D}, \mathcal{B}), \tag{1}$$

Table 1 Questions that turn a Physical AGI aspiration into an evidence plan. These are proposed reporting dimensions, not thresholds that certify AGI.

Dimension	Question	Evidence to report
Grounded competence	Can decisions produce the intended physical outcome?	Closed-loop task success and outcome verification, not descriptions alone.
Breadth and transfer	What remains effective outside familiar conditions?	Separate held-out tasks, objects, scenes, dynamics, and embodiments.
Adaptation and retention	What can be learned from limited new experience?	Performance versus adaptation budget; retests of earlier skills.
Autonomy and recovery	When is external help required?	Intervention counts, recovery outcomes, and elapsed execution time.
Safety and reliability	Which failures occur and with what consequences?	Constraint violations, failure severity, uncertainty, and safe-stop behavior.

where the components index tasks, objects, environments, dynamics, and embodiments. This is reporting notation, not an assumption that these factors are independent. A split may withhold new instances of a familiar object, unfamiliar task compositions, or a new embodiment; these are different forms of novelty. Authors should identify which factors are held out, which remain shared, and what overlap with pretraining can actually be audited. Merely renaming instructions does not establish a held-out task family.

Adaptation must have a declared cost. Report both performance before adaptation and performance as a function of an explicit budget,

$$b = (n_{\text{demo}}, n_{\text{trial}}, c_{\text{adapt}}, n_{\text{assist}}), \quad (2)$$

covering demonstrations, environment trials, adaptation compute, and human assistance. These quantities are not interchangeable and should not be collapsed into one score without a justified weighting. Resetting an environment, selecting successful trials, and supplying corrective instructions also consume resources that should be disclosed. Retesting previously acquired skills distinguishes cumulative learning from improvement bought by forgetting. In this framing, progress is a reproducible shift in breadth and learning efficiency under stated conditions, not an unqualified claim that a system is general.

3 Language Priors for Reusable Physical Intelligence

LLM-based world knowledge refers to language-mediated world priors stored as parametric regularities in large language models (Petroni et al., 2019; Roberts et al., 2020). Prompts and context expose these priors as task hypotheses about objects, goals, actions, procedures, and constraints for Physical AI (Brown et al., 2020; Wei et al., 2022). The attraction of this starting point is reuse: an agent need not rediscover every object function or task procedure through physical trials. For progress towards Physical AGI, however, a prior matters only insofar as it helps the agent acquire or adapt competence when the task or situation changes. A familiar instruction such as opening a drawer illustrates the distinction: knowing the procedure may narrow the search, but an unfamiliar drawer latch can still require observation, experimentation, and revision.

3.1 World Knowledge as Coupled Priors.

World knowledge is better treated as a set of coupled priors than as a monolithic store: semantic knowledge grounds object categories and references; commonsense knowledge supplies default assumptions and safety constraints; procedural knowledge decomposes tasks; causal knowledge suggests outcomes and risks; spatial knowledge supports layout and manipulation preconditions; and affordance knowledge links objects or parts to possible interactions (Arkin, 1990; Hagoort et al., 2004; Schwenk et al., 2022). In grounded Physical AI systems, spatial and affordance priors are often operationalized by models that connect linguistic relations

to perceptual structure: recent spatial VLMs map relations such as “inside” or “aligned with” to metric or 3D structure (Chen et al., 2024; Yuan et al., 2025; Song et al., 2025). Affordance-oriented work similarly identifies which objects, parts, or regions support a requested action (Qian et al., 2024; Huang et al., 2024; Chu et al., 2025).

3.2 Parametric World Knowledge in LLMs.

LLMs encode factual associations, procedural patterns, commonsense defaults, and task constraints through large-scale pretraining on corpora that repeatedly describe object functions, task procedures, safety constraints, and likely outcomes (Petroni et al., 2019; Roberts et al., 2020; Brown et al., 2020; Bommasani et al., 2021). Because these descriptions recur across documents, they become parametric regularities whose strength tracks pretraining exposure (Kandpal et al., 2023; Chang et al., 2024). Recent recall and materialization studies further probe which factual priors can be surfaced from LLMs (Yuan et al., 2024; Hu et al., 2025). Embodied planning work provides a task-level diagnostic: LLMs can often retrieve plausible steps, object uses, and constraints for situated tasks (Song et al., 2023; Rana et al., 2023; Hua et al., 2025).

3.3 LLMs as Priors and Controllers in Physical AI.

LLMs contribute to Physical AI in four main ways. For *task planning*, SayCan (Ichter et al., 2023) grounds LLM-generated step sequences with learned affordance values to filter physically infeasible actions, and Inner Monologue (Huang et al., 2023c) enables closed-loop replanning by feeding back environmental observations as language (Huang et al., 2022). Moving from prose to code, *skill generation* approaches such as Code as Policies (Liang et al., 2023) and ProgPrompt (Singh et al., 2023) prompt LLMs to produce executable robot programs, while Voyager (Wang et al., 2023) uses LLMs to iteratively build open-ended skill libraries. At the level of *goal and reward specification*, VoxPoser (Huang et al., 2023a) composes spatial value maps from language instructions, reducing manual reward engineering. Finally, as *agentic orchestrators*, LLMs coordinate tools and sub-modules across multi-step tasks (Rana et al., 2023; Hua et al., 2025), acting as closed-loop decision engines rather than one-shot planners. Together, these uses show how language models can supply reusable semantic and procedural structure to embodied systems. Their contribution should be judged by what this structure enables an agent to learn and execute, alongside the perception, skills, and tools supplied by the rest of the system.

3.4 Limits of Language-Only Physical Reasoning.

The systems discussed above reveal a common pattern: LLMs can propose useful goals, plans, and constraints, while grounding, verification, and action interfaces help connect these proposals to feasible behavior (Ichter et al., 2023; Huang et al., 2023c; Rana et al., 2023). The core limitation is the abstraction gap between language and physics. Language compresses continuous physical states into sparse descriptions and often omits pose, velocity, contact, deformation, uncertainty, and embodiment constraints. Knowing that glass is fragile or that a handle can be grasped does not by itself establish a calibrated estimate of the geometry, contact, force, friction, or trajectory of a particular interaction; physical-reasoning evaluations probe aspects of this gap (Qiu et al., 2025; Xu et al., 2025). LLMs can therefore hallucinate objects, propose infeasible plans, or assume that actions succeed. Recent planning analyses, including LLM-modulo studies and PlanBench updates, document reliability limits in evaluated language-model planners and motivate combining proposals with explicit verification (Kambhampati et al., 2024; Valmeekam et al., 2024). Physical reasoning benchmarks further expose weaknesses in dynamics, vision-grounded physics, and tool use (Xiang et al., 2025; Zhang et al., 2025c). The question carried forward is therefore concrete: can a language prior be revised when observation contradicts it, and can that revision improve the next attempt? Perceptual grounding supplies the evidence for this revision; action and prediction determine whether the revised belief is useful. Language models offer one route to this process, without defining a necessary architecture for Physical AGI.

4 Grounding World Knowledge into Perception

To use language-based world knowledge in the physical world, an agent must connect it to perceptual observations. VLMs and MLLMs offer a major bridge in this transition: they align language with images,

videos, regions, objects, and scenes (Radford et al., 2021; Jia et al., 2021; Alayrac et al., 2022; Liu et al., 2023c). Language-only knowledge is underspecified with respect to the current world state: an LLM may know that glass is fragile or that a handle affords pulling, but an embodied system must determine whether the relevant object is present, where it is, and whether the current configuration makes that prior actionable. Thus, perception makes world knowledge situated, task-relevant, and actionable.

4.1 Vision-Language and Multimodal Models.

The development of VLMs can be viewed as a progression from image-text representation alignment to multimodal instruction following. Contrastive and cross-modal pretraining made language an open-vocabulary interface for visual concepts, allowing models to retrieve, classify, and reason over images beyond fixed label spaces (Radford et al., 2021; Lu et al., 2019; Tan and Bansal, 2019). More recent MLLMs connect visual encoders to large language backbones through cross-attention, query transformers, projection layers, or instruction tuning, enabling few-shot visual reasoning, visual dialogue, and open-ended image understanding (Team, 2026; Li et al., 2023b; Shen et al., 2026c; Li et al., 2026; Shen et al., 2026d). For Physical AI, their importance is that instructions, observations, and task context can be interpreted through a shared language interface. A VLM can identify task-relevant objects and connect them to commonsense priors, making it useful as a high-level perceptual reasoner in open-world environments. Yet this grounding remains mostly semantic rather than physical: VLMs do not automatically recover metric state, physical parameters, or control-relevant uncertainty (Agarwal et al., 2025; Liu et al., 2025a; Shridhar et al., 2022; Bear et al., 2021).

4.2 Spatial, Temporal, and Affordance Grounding.

Physical AI requires grounding beyond object categories: agents must localize objects, resolve spatial relations, and infer action affordances. Recent MLLMs provide stronger grounded outputs through region, mask, and point supervision (Rasheed et al., 2024; Lai et al., 2024), unified detection-grounding-segmentation (Xiao et al., 2024), and explicit pointing supervision (Deitke et al., 2025). For robotics, recent work extends grounding toward robotics-relevant spatial reasoning and language-conditioned affordance prediction (Chen et al., 2024; Song et al., 2025; Pothiraj et al., 2025). These works improve perceptual grounding, but Physical AI still requires metric state, reachability, uncertainty, and embodiment-specific constraints. Temporal grounding moves perception from static recognition to event localization and state change (Shen et al., 2026e). Recent video MLLMs explicitly model timestamps and grounded moments, including TimeChat Ren et al. (2024), VTG-LLM Guo et al. (2025), Grounded-VideoLLM Wang et al. (2024), VQToken Zhang and Fu (2026), and TimeSuite (Zeng et al., 2025); VideoGLaMM further extends grounding to pixel-level video regions (Munasinghe et al., 2025). However, temporal grounding is still not physical dynamics: Video-MME-v2 and PhysBench show that long-video understanding and physical-world reasoning remain difficult for current MLLMs (Fu et al., 2026a; Chow et al., 2025). Affordance grounding connects perception to possible actions. Recent work uses VLM/LLM priors to predict interaction regions, 3D affordances, or language-conditioned affordance points, as in AffordanceLLM, PAVLM, Palm, and RoboPoint (Qian et al., 2024; Liu et al., 2025b, 2026; Yuan et al., 2025). These models make perception more action-relevant, but feasibility still depends on embodiment, geometry, reachability, & low-level control.

4.3 From Semantic Grounding to Control-Relevant Evidence.

The bottleneck arises when an interface reduces a rich observation to a sparse description before control or prediction can use it. Text can encode coordinates, uncertainties, and structured state, and multimodal architectures can expose dense features; neither an output modality nor a model label determines physical competence. The relevant issue is whether the representation preserves the quantities required by the task and whether its errors are measured. Existing evaluations probe complementary aspects: BLINK examines low-level visual perception (Fu et al., 2024), Video-MME tests long-horizon temporal understanding (Fu et al., 2025), and PhysBench, QuantiPhy, and MASS-Bench evaluate physical reasoning, quantitative physics, and motion-aware spatiotemporal grounding (Chow et al., 2025; Puyin et al., 2025; Wu et al., 2025).

For the drawer example, recognizing its handle or latch is only an initial observation. The agent may need to change viewpoint, make a cautious contact, or compare expected motion with tactile and proprioceptive feedback. This motivates an active-perception criterion for progress towards Physical AGI: can the agent

choose an observation or interaction that resolves uncertainty relevant to an unfamiliar task? Touch, force, and proprioception belong in this criterion whenever vision leaves the relevant state ambiguous. Grounding then becomes a process of acquiring evidence, whose value is determined by the action decision it improves.

5 Grounding World Knowledge into Action

Perceptual grounding identifies evidence relevant to a task; action tests whether an agent can use that evidence to change the world. Vision-Language-Action (VLA) models connect this transition by mapping visual observations, language goals, and, where supported, embodiment states into action outputs. The action interface is consequential: it determines which aspects of prior knowledge can be executed, which must be learned from interaction, and which are tied to a particular body. We examine this interface through action representations, generalist policy learning, and the coupling of reasoning with control.

5.1 Action Spaces and Policy Learning.

The action space defines how a Physical AI system acts. Depending on the embodiment and task, actions can be represented as poses, joint states, gripper commands, contact targets, waypoints, trajectories, action chunks, or high-level skills. This action representation determines how pretrained world knowledge becomes executable behavior: high-level skills align better with language and planning, while low-level commands provide precise but embodiment-specific control.

Recent VLA systems instantiate this spectrum through several action interfaces. One line discretizes robot actions into tokens, enabling autoregressive prediction with transformer-based VLM backbones, as in RT-2 [Zitkovich et al. \(2023\)](#) and OpenVLA [Kim et al. \(2025\)](#); FAST further improves this direction by tokenizing high-frequency action sequences in frequency space, and can be combined with models such as π_0 to form efficient autoregressive VLA policies [Pertsch et al. \(2025\)](#). A second line predicts continuous action chunks or trajectories, following action-chunking and diffusion or flow-based visuomotor policies such as ACT [Zhao et al. \(2023\)](#), π_0 [Black et al. \(2025\)](#), $\pi_{0.5}$ [Physical Intelligence et al. \(2025b\)](#), DexVLA [Wen et al. \(2025a\)](#), and RDT-1B [Liu et al. \(2025c\)](#). A third line spatially structures the action interface beyond raw motor commands. SpatialVLA [Qu et al. \(2025\)](#) represents actions with adaptive 3D action grids, grounding candidate motions in egocentric coordinates. 3D-VLA [Zhen et al. \(2024\)](#) uses 3D interaction tokens to predict task-relevant targets or motion primitives in scene space. Such spatial action representations expose geometry to the policy and can improve transfer across objects, scenes, and embodiments.

5.2 Grounding Language in Action.

Early VLAs asked whether a VLM could act by treating actions like language. PaLM-E showed that embodied multimodal representations can support robot reasoning and planning [Driess et al. \(2023\)](#). RT-2 made the link to control explicit by co-fine-tuning a pretrained VLM on web-scale vision-language tasks and robot trajectories, representing robot actions as text-like tokens, and de-tokenizing outputs back into commands [Zitkovich et al. \(2023\)](#). This provides a concrete route for aligning web-trained representations with robot action data. In the related setting of human motion generation, FineMoLA studies fine-grained motion-language alignment ([Wang et al., 2026](#)); this is a representational connection, not evidence of robot-policy transfer. Scaling this recipe requires broader data and transferable policy interfaces. Open X-Embodiment and RT-X standardize data across many robot embodiments [O’Neill et al. \(2024\)](#). Octo learns an open generalist policy from heterogeneous data [Octo Model Team et al. \(2024\)](#), and OpenVLA trains on large-scale real robot demonstrations [Kim et al. \(2025\)](#). Their observation interfaces and action heads differ: OpenVLA uses discrete action tokens, whereas Octo uses diffusion-based action prediction. This diversity matters when comparing transfer claims, because available state inputs, control frequency, robot hardware, and adaptation procedures can all change the difficulty of a task. Training across many tasks and embodiments broadens experience, but does not alone establish learning efficiency on a held-out task or body.

5.3 Reasoning to Action.

Recent systems increasingly use hybrid VLA architectures that separate world knowledge and reasoning from motion generation. A VLM grounds instructions, object semantics, spatial relations, task history, and commonsense priors, while a specialized policy converts this reasoning into high-frequency actions under embodiment constraints. This shifts VLA from action-as-language toward a reasoning-to-control interface for Physical AI. π_0 combines a pretrained VLM with a flow-matching action expert Black et al. (2025), and $\pi_{0.5}$ adds heterogeneous co-training over robot data, web data, and semantic prediction tasks Physical Intelligence et al. (2025b). DexVLA uses a VLM with a diffusion expert Wen et al. (2025a), RDT-1B scales diffusion for bimanual manipulation Liu et al. (2025c), and GR00T N1 combines vision-language reasoning with a diffusion transformer action generator for humanoids NVIDIA et al. (2025). TinyVLA Wen et al. (2025b), SmolVLA Shukor et al. (2025), Xiaomi-Robotics-0 Cai et al. (2026), and StarVLA- α Ye et al. (2026) show that efficiency and real-time execution are also part of the roadmap. The trend is also moving beyond offline imitation. $\pi_{0.6}^*$ uses reinforcement learning from real deployments with experience and corrective interventions Physical Intelligence et al. (2025a), while MEM adds video and text memory for longer-horizon behavior and adaptation from history Torne et al. (2026).

5.4 From Trained Repertoires to Reusable Competence.

VLA is an interface description, not a claim about the breadth of intelligence or the presence of predictive internal representations. The question for Physical AGI is how much of a policy’s competence survives a change in task composition, object dynamics, or embodiment, and how much experience is required to repair what does not. For an unfamiliar drawer, a useful transfer might preserve the goal and the ordering of subskills while adapting the contact motion; on a different robot, the same goal may require a different grasp and trajectory. Evaluation should distinguish such reuse from training anew on each target, and report demonstrations, interaction, interventions, and computation used for adaptation.

The memory and deployment-learning examples above suggest complementary routes beyond a fixed repertoire: retain informative failures, retrieve relevant experience, and update behavior when evidence changes. Each route requires checking whether a newly acquired skill composes with earlier ones and whether earlier competence is retained. A predictive model offers another route by estimating candidate action consequences before execution. The next question is therefore whether prediction improves decisions under novelty, including recognizing when its own forecasts are too uncertain to justify an action.

6 World Models for Prediction and Adaptation

The previous sections describe how LLMs provide high-level world knowledge, how VLMs ground such knowledge into perception, and how VLAs connect perception and language to executable actions. Predicting an action from experience, however, does not by itself demonstrate an ability to compare alternative futures under unfamiliar conditions. Policies may encode dynamics implicitly, and some systems combine policy and predictive objectives; the VLA label alone does not determine whether a separate world-model component is present or needed. Explicit *world models* offer one way to expose predictive knowledge learned from videos, trajectories, and embodied interactions so that it can be used and evaluated in decision making.

We use *world model* to refer to a model that predicts or simulates future observations, latent states, rewards, values, or action consequences from current states and possible actions. This is a functional distinction from language-mediated world knowledge, rather than a strict division between model families. Language priors can suggest plausible events and actions; a predictive model used for control must relate a specified state and intervention to measurable consequences. A single architecture may serve both functions, and a useful predictive representation need not reconstruct every physical variable. This idea has a long history in model-based learning and control: early neural world models learned compressed spatiotemporal representations for agents (Ha and Schmidhuber, 2018), latent dynamics models such as PlaNet and Dreamer learned to plan and act through latent imagination (Hafner et al., 2019, 2020, 2023), and MuZero showed that decision-centric models can support planning by predicting value, reward, and policy-relevant quantities without explicitly reconstructing observations (Schrittwieser et al., 2020). For Physical AI, recent world models can

be organized by their prediction target and interaction interface. A more detailed taxonomy with additional representative works is provided in Appendix A.4.

6.1 Video-space World Models.

Video-space world models generate or predict future visual observations, providing an intuitive interface for visual imagination, synthetic data generation, and future scene simulation. Representative examples include generative driving and real-world simulators such as GAIA-1 and UniSim, as well as interactive environment models such as Genie (Hu et al., 2023; Yang et al., 2024; Bruce et al., 2024). Cosmos further frames world foundation models as general-purpose world models that can be adapted to Physical AI applications such as robotics, autonomous driving, and synthetic data generation (Agarwal et al., 2025). However, photorealistic video generation alone is not sufficient for Physical AI. For comparing candidate actions, a video-space model should predict temporally consistent, action-conditioned consequences with physically plausible dynamics. Passive video modeling may still support representation learning without satisfying that decision-making requirement.

6.2 Latent World Models.

Latent world models predict future states in representation space rather than reconstructing pixels. This is important because dense video generation can be computationally expensive and may allocate capacity to visual details that are irrelevant to control. Latent models can target task-relevant dynamics, making them attractive for efficient planning, policy learning, and long-horizon imagination, although this depends on what their learning objectives preserve. Beyond latent dynamics in model-based reinforcement learning, JEPA-style predictive architectures argue for learning world models directly in representation space (LeCun, 2022; Assran et al., 2023; Bardes et al., 2024). Recent V-JEPA 2 further connects self-supervised video representation learning with robot planning by post-training an action-conditioned latent world model from robot trajectories (Assran et al., 2025).

6.3 Interactive and Action-conditioned World Models.

Interactive and action-conditioned world models connect prediction directly to embodied decisions by representing futures associated with possible actions (Shen et al., 2026b). Rather than passively predicting what may happen next, these models estimate what would happen if a particular action were taken. This capability supports planning, policy learning, simulation-based training, safety evaluation, and recovery. It also clarifies a possible division of work: a policy proposes actions, while a predictive model estimates their consequences; a joint model can implement both functions. For progress towards Physical AGI, these consequences must remain useful when familiar appearances conceal changed mechanics or when familiar skills are composed in a new order. The drawer example makes the requirement concrete: a visually plausible forecast is insufficient if it predicts that a jammed drawer opens under an ineffective pull.

6.4 Decision Value under Unfamiliar Dynamics.

We propose assessing predictive competence through three linked questions: whether forecasts distinguish candidate actions, whether errors increase detectably outside familiar conditions, and whether new observations enable useful adaptation. The comparison should hold the policy, interaction budget, and planning computation sufficiently controlled to attribute an improvement to prediction rather than additional resources. Useful outcomes include improved task completion, earlier detection of unsafe choices, or fewer interactions needed to learn a changed mechanism; none follows automatically from visual quality or a low average prediction error. At deployment, accumulated prediction error also requires feedback: the agent must compare forecasts with observations, revise its belief, and replan or defer when necessary. This closes the conceptual path from a reusable language prior to an integrated agent that can test, update, and act on that prior.

7 From Capable Models to Integrated Physical Agents

A useful plan, a feasible action, and a plausible prediction must eventually agree in one unfolding interaction. The system-level challenge is to maintain that agreement as observations change and actions fail. This section connects the earlier interfaces to execution, memory, and feedback, while separating present deployment capabilities from broader claims of generality.

7.1 Embodied Agents.

Embodied agents connect high level task understanding with physical execution. A typical system may parse an instruction, infer task structure, ground objects and states, choose actions and trajectories, execute them through policies or controllers, and verify whether the expected outcome has been achieved (Guo et al., 2026; Ichter et al., 2023; Liu et al.; Team et al., 2025b,a; Liu et al., 2023a; Fu et al., 2026b). Tracking, layout inference, and trajectory prediction under noisy or occluded observations address supporting state-estimation problems rather than demonstrating that entire execution loop (Zhang et al., 2026b, 2024a, 2023). This differs from a pure VLA formulation because the action model is only one component of the full system. System performance also depends on state estimation, execution interfaces, controller behavior, and online verification. Many embodied agents therefore use modular interfaces between reasoning and control. High level modules may operate over language, symbolic states, object relations, keypoint constraints, value maps, or robot programs, while low level modules execute motion primitives, grasp planners, visuomotor policies, or controllers (Huang et al., 2025, 2023a; Liang et al., 2023; Shen et al., 2026a). This modularity can make long horizon behavior easier to specify and diagnose, but it also raises interface design questions. The system must decide what information should be passed between modules, how uncertainty should be handled, and when the agent should replan instead of continuing execution.

7.2 Deployment and System Integration.

A central challenge in Physical AI is the gap between offline evaluation and interactive deployment. Offline evaluation measures whether a model matches reference labels on fixed data, whereas deployment tests whether these predictions can drive closed-loop progress in the physical world. Open-loop accuracy can hide compounding errors, poor recovery, and sensitivity to state deviations, so robust deployment requires systems to observe the environment, execute actions, verify outcomes, and update plans when reality differs from expectation. Recent work explores this direction through real-world agent platforms (Guo et al., 2026), open-vocabulary planning and grounded decoding (Huang et al., 2023b), LLM/VLM-based robot decision frameworks (Jin et al., 2024), and deployment-oriented action reasoning models (Shen et al., 2026a; Fang et al., 2026). Another key design choice is the interface between learned models and robot controllers. Frontier models may produce language plans, symbolic predicates, object constraints, value maps, or robot code, while robots execute controller references. An effective interface should be expressive enough to support diverse tasks while remaining constrained enough for stable execution. This motivates systems that connect learned perception and reasoning to symbolic world models (Athalye et al., 2026), task-and-motion planning (Kumar et al., 2026), optimization and constraint-based motion generation (Huang et al., 2025, 2023a; Liu et al.), and robot-program or policy-level execution interfaces (Wu et al., 2023; Zhang et al., 2024b).

7.3 Persistent Experience and Corrective Feedback.

Returning to the unfamiliar drawer, successful adaptation should leave something reusable: a remembered obstruction, a revised belief about friction, or a skill that transfers to another drawer. These are different kinds of updates. Episodic memory records what happened; state estimation tracks what is currently true; policy or model learning changes how future interactions are predicted or controlled. A long context window or a transcript is not, by itself, evidence that the agent retains and generalizes a physical skill.

We therefore treat the experience loop as an experimental object. An integrated agent should expose when it updates state, changes a plan, stores a useful experience, or modifies a skill. A trial in which it improves after a correction should be followed by an unassisted retest and, where relevant, a transfer test. This makes the contribution of memory and learning distinguishable from extra human instruction. The appropriate update mechanism can be modular, end-to-end, or hybrid; its value lies in measurable changes to future behavior.

7.4 Frontier Digital Agents: What Transfers, and What Does Not.

Frontier reasoning models make the distinction between digital agency and physical competence especially important. OpenAI’s official documentation describes GPT-6 Astra as a model for complex reasoning, coding, computer use, research, and document creation; it accepts text and image inputs and supports tool use (OpenAI, n.d.). We cite this as product documentation, not as a robotics paper or an embodied evaluation. These capabilities suggest potential roles in task decomposition, tool orchestration, and program synthesis, but those roles remain hypotheses to test in a physical system.

The relevant experiment is not whether a frontier model can narrate a robot task. It is whether replacing the reasoning component improves execution or adaptation when observations, controllers, tools, and assistance budgets are held fixed. Any comparison should identify the model version, available tools, action interface, latency, memory, and human support. Performance in computer-use workflows does not supply evidence about contact dynamics, cross-embodiment transfer, or physical recovery. Demonstrating those properties requires closed-loop embodied tests of the integrated agent.

8 From Benchmark Coverage to Evidence of Generality

Evaluation remains a central challenge for Physical AI. Static language and vision benchmarks measure recognition, reasoning, or prediction under fixed inputs, whereas claims about embodied generality require evaluation under interaction, where actions change future observations and outcomes. Benchmarks should therefore test whether systems can ground knowledge into perception, planning, execution, and recovery, rather than only answer questions or predict actions offline. Existing efforts study embodied task execution in simulation and household environments (Srivastava et al.; Li et al., 2023a; Li et al.; Shen et al., 2021), embodied reasoning and agent evaluation (Li et al., 2024; Yang et al., 2025; Shen et al., 2026f), and embodied spatial intelligence with closed perception-action loops (Hong et al., 2026). Closed-loop execution is especially important because open-loop prediction cannot fully capture compounding error, contact dynamics, safety failures, physical constraints, or recovery behavior. Benchmarks should evaluate task completion, failure modes, intervention counts, robustness to perturbations, and generalization across objects, scenes, instructions, initial states, and embodiments. Recent efforts support cross-task policy learning (Liu et al., 2023b; Fei et al., 2025; Zhou et al., 2025), deployment-oriented action evaluation (Fang et al., 2026), perception-action loop evaluation (Hong et al., 2026), physical reasoning (Huang et al., 2026), large-scale simulation and household evaluation (Zhu et al., 2020; Nasiriany et al., 2024, 2026; Yang et al., 2026), and automated task and data generation for sim-to-real policy learning (Gong et al., 2025). For claims about deployment on physical robots, real-robot evaluation remains necessary because implementation details can substantially affect performance. Evidence of embodied generality therefore requires system-centric evaluation, comparing what agents can reliably do under interaction, not only what their components predict from static inputs.

Benchmark coverage as a starting point. An agent that has mastered one kitchen may still need to learn how an unfamiliar drawer opens, remember where a person moved a tool, or recover when a grasp fails. Table 2 makes the scale of existing evaluation resources concrete while distinguishing their counting units. These resources expose complementary parts of the problem: transfer between manipulation tasks, the breadth of household activities, composition in varied scenes, and the gap between high-level planning and low-level action. Their coverage provides a foundation for evaluating generality; evidence towards Physical AGI additionally requires held-out task families and explicit limits on adaptation experience, together with measurements of retention, recovery, and human intervention.

8.1 A Protocol for Testing Transfer and Adaptation.

The dimensions in Section 2 can be evaluated with a staged protocol. First, freeze the pretrained system, its tools, and its controller interfaces, and report performance on a familiar reference distribution. Next, evaluate declared held-out factors without task-specific updates. Then allow adaptation under the budget b and report the resulting learning curve rather than only its endpoint. Finally, retest previous skills and evaluate a further variation that was not used for adaptation. This sequence distinguishes immediate transfer, acquisition, retention, and transfer of newly acquired knowledge.

Table 2 Reported coverage of four embodied evaluation resources, using the versions described in their original papers. Counts describe different units and do not form a ranking of difficulty or agent capability. The final column summarizes each resource’s diagnostic focus.

Resource	Reported scale	Coverage structure	Diagnostic focus
LIBERO (Liu et al., 2023b)	130 manipulation tasks	4 suites: 10 + 10 + 10 + 100 tasks	Transfer of spatial, object, goal, and combined knowledge
BEHAVIOR-1K (Li et al., 2023a)	1,000 activity definitions	50 scenes; logical initial and goal conditions	Everyday activity breadth and long-horizon physical interaction
RoboCasa (2024) (Nasiriany et al., 2024)	100 task definitions	25 atomic + 75 composite tasks; 120 kitchen scenes	Skill composition and variation in kitchen environments
EmbodiedBench (Yang et al., 2025)	1,128 testing instances	4 environments; 6 capability subsets	High-level planning through low-level navigation and manipulation

Counting note. LIBERO uses the original four-suite design; the 100-task suite is further partitioned in some protocols. BEHAVIOR-1K counts activity definitions, and RoboCasa counts designed tasks rather than evaluation episodes. EmbodiedBench counts testing instances. Scene counts and environment counts refer to different levels of organization. These figures are paper-reported coverage, not results from a new evaluation.

For each stage, report the number of tasks, episodes, initializations, and independent training runs where applicable, together with uncertainty estimates and the unit over which they are computed. Episodes from the same task or training run are not necessarily independent. Human resets, retries, privileged simulator state, manually selected prompts, and successful-trial filtering should be explicit. When training exposure is undisclosed, the result can establish performance on a test resource but not verified novelty relative to all prior experience.

8.2 Comparisons that Localize the Source of Progress.

A stronger Physical AGI narrative requires stronger attribution. To test the contribution of a language prior, compare it with an appropriately matched goal or planning interface. To test a predictive model, compare actions selected with and without its predictions under a matched total resource budget. To test memory, evaluate retention and transfer while controlling access to past observations and corrective feedback. These are proposed controls, not experiments conducted in this paper.

Aggregate task success should be accompanied by breakdowns over held-out factors and by operational outcomes such as intervention count, constraint violations, completion time, and recovery success conditional on a specified failure. Those outcomes reveal whether improvement is broad or concentrated in a familiar subset. Current benchmark coverage is valuable infrastructure, but neither a large task count nor a single leaderboard position settles whether an agent learns efficiently across physical domains.

9 Research Gaps towards Physical AGI

The roadmap becomes a research program only when its interfaces lead to testable improvements in generality. We identify six gaps spanning cognition, physical interaction, and adaptation. These are prospective criteria for progress towards Physical AGI, not capabilities established by combining the model families discussed above; additional technical challenges are discussed in Appendix A.8.

9.1 Acquiring Unfamiliar Tasks from Reusable Priors.

LLMs provide a store of language-mediated associations that can inform a new task (Petroni et al., 2019; Roberts et al., 2020). The unresolved issue is how this knowledge supports problem formulation, causal reasoning, and skill acquisition when the required procedure was not supplied in training. A useful milestone is learning a held-out task family with a declared amount of instruction, demonstration, interaction, and computation, then applying that knowledge to a further variation. Novel combinations should require dependencies between skills, so that success cannot be explained by independently replaying familiar routines. Evidence should include learning curves and failure cases, alongside the final success rate.

9.2 Seeking Physical Evidence under Uncertainty.

Vision-language alignment connects semantic priors to observations (Radford et al., 2021; Liu et al., 2023c), but an unfamiliar interaction may depend on quantities that the current view does not reveal. An agent should determine when to inspect, reposition a camera, probe with a tool, or use touch and proprioception to resolve that uncertainty. The research challenge is to couple information gathering with task costs and safe interaction constraints. A useful milestone is improved performance under occlusion or changed material properties with the sensing and interaction budget held explicit. This tests whether additional perception changes a decision, rather than merely enriching a verbal description.

9.3 Transferring Skills across Bodies and Dynamics.

VLA systems connect language and perception to executable actions through learned interfaces (Zitkovich et al., 2023; Kim et al., 2025). General transfer requires deciding what should remain invariant, such as a goal or contact relation, and what must adapt, such as reachability, kinematics, force, and timing. Held-out embodiments and altered dynamics should therefore be evaluated separately from unseen appearances, with any controller engineering or additional training disclosed. A useful milestone is reusing a skill on a new body with bounded adaptation while maintaining the original skill on its source body. This criterion allows embodiment-specific controllers while making their contribution to apparent generality visible.

9.4 Retaining and Composing Experience.

Long-lived agents need to preserve useful experience across episodes and changing tasks. We distinguish episodic records of particular interactions, semantic knowledge that abstracts across them, and procedural knowledge that supports execution; the implementation may use retrieval, parameter updates, external storage, or combinations of these mechanisms. The key issue is whether acquired knowledge remains usable after subsequent learning and can be composed to solve a new problem. A useful milestone is sustained acquisition and retention across a sequence of task families, with memory capacity, retrieval cost, and update data reported. Repeating tasks after intervening learning is necessary to detect forgetting that an isolated final-task score would conceal.

9.5 Predicting Consequences that Improve Decisions.

World-model research offers predictive and simulative mechanisms spanning compact latent dynamics and generated observations (Ha and Schmidhuber, 2018; Agarwal et al., 2025; Lin et al., 2026). For embodied use, the open question is when their predictions are sufficiently action-sensitive, physically informative, and efficient to change a decision beneficially. A useful milestone is a reduction in failed or unnecessary interactions on unfamiliar dynamics under a matched total resource budget. Tests should also expose when model error makes planning worse, and whether the agent can detect that condition and return to observation or a conservative policy. This makes calibration and recovery part of predictive competence.

9.6 Sustaining Reliable Autonomy with Transparent Evidence.

Deployment connects perception and action through a closed loop, making integrated evaluation essential (Yang et al., 2025). Progress should be measured over extended episodes with disturbances, delayed feedback, failed actions, and opportunities to request help or stop. A useful milestone combines broad task

completion with explicit rates of intervention, recovery, and safety-constraint violation, evaluated on disclosed held-out conditions. Autonomy must be reported separately from task breadth: a system supported by frequent human correction can still be scientifically valuable, but its assistance budget is part of the result. Comparable evidence also requires disclosing training exposure where known and marking uncertainty where contamination or access restrictions prevent that assessment.

10 Conclusion

The path from language priors towards Physical AGI is a question of how reusable knowledge becomes competence under unfamiliar physical conditions. Our analysis traces this path through perceptual evidence, executable actions, predictive models, and integrated agents. These functions can be distributed across modules or learned within a shared architecture; assembling them does not by itself establish general intelligence. The stronger test is whether an agent can acquire a new task, adapt to changed dynamics or embodiment, retain what it learns, and recover from failure within an explicit experience and assistance budget.

Physical AI names a broad class of systems; Physical AGI remains an aspirational regime of general cognitive and physical capability. Language provides a promising source of transferable priors, while interaction supplies evidence that can confirm or revise them. The proposed roadmap connects these two sources of knowledge and identifies where controlled evaluation can establish progress. Its central criterion is reusable intelligence demonstrated through learning and transfer, alongside the reliability required for action in the physical world.

11 Limitations

This article develops a conceptual synthesis and research framework from existing literature; it introduces no new model, training method, or empirical benchmark result. Its language-centered starting point gives selective coverage to LLMs, multimodal grounding, action interfaces, world models, and integrated agents. Classical control, morphology, tactile and audio perception, materials, deformable-object dynamics, and domain-specific engineering receive less detailed treatment despite their relevance to physical competence. The proposed criteria are not a validated test or threshold for Physical AGI, and the work does not establish that any system has achieved it. Nor does the framework imply that language models, explicit world models, or a particular modular arrangement are necessary for general intelligence. Reported results from different publications can vary in training exposure, hardware, tasks, and intervention budgets; they should not be read as a controlled ranking unless their protocols support that comparison.

12 Broader Impact

The intended contribution is to make research claims about physical intelligence more precise and comparable. Separating task breadth, learning efficiency, retention, and autonomy can help readers distinguish progress on a component from evidence about an integrated agent. The Physical AGI framing also creates a risk of overstating what existing demonstrations establish; the aspiration should therefore remain tied to explicit evaluation conditions and disclosed limitations.

Systems that act in the world can affect people through physical error, inappropriate goals, or failures to recognize uncertainty. The framework consequently treats intervention, recovery, and safe stopping as observable properties alongside successful task completion. It also calls for transparent accounts of human assistance and resource use, which influence both scientific comparisons and practical accessibility. This work releases no new model, dataset, robot policy, controller, or deployment system; its direct contribution is analysis and research guidance.

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Appendix

A Supplementary Material

This appendix documents the scope, technical categories, and supporting literature behind the conceptual framework. It expands the connections from language priors to perception, action, prediction, and integrated Physical AI systems, then examines how these connections inform research towards Physical AGI. The tables identify representative interfaces and evaluation questions; they provide neither an exhaustive catalogue nor a ranking of methods.

A.1 Scope and Boundary

The framework selects literature according to its contribution to the use of prior knowledge in physical perception, prediction, planning, learning, and action. This includes spatial and temporal grounding, language-conditioned policies, and predictive representations used for decisions. It gives less detailed coverage to classical control, hardware design, sensing, and domain-specific engineering, although these remain important to physical competence.

Table 3 makes this editorial boundary explicit. The language-centered starting point defines the paper’s emphasis, not membership in Physical AI: a system can perceive, predict, reason, or act in a physical environment without an LLM. Likewise, the Physical AGI aspiration concerns the breadth and efficiency of learning and transfer, rather than adherence to the roadmap’s architectural categories.

The table should be read as a boundary rather than a separation. Many excluded areas remain important to Physical AI, but they are not the organizing focus of this paper. For instance, low-level manipulation control and hardware design are indispensable for deployment, yet our discussion emphasizes how foundation models and world models interface with such systems. Likewise, video generation is relevant when it becomes temporally consistent, controllable, and action-conditioned, but generic video synthesis is not equivalent to physical world modeling.

A.2 Connections to Existing Research Perspectives

Different research perspectives foreground different questions: deployment connects intelligence to sensing and infrastructure; visual modeling examines representation and simulation; policy learning connects goals to control; predictive modeling examines possible consequences. Table 4 relates these questions to the present framework without assigning coverage scores or relative superiority.

Our organizing question is how prior knowledge becomes reusable competence under unfamiliar physical conditions. Language is one source of that knowledge. Its usefulness depends on the observation and action interfaces through which an agent tests and revises a prior; a sparse language interface may omit control-relevant quantities even when the underlying model contains useful physical information.

The roadmap follows dependencies among functions while allowing feedback and shared implementations. Observations can revise a prior, actions can seek information, and prediction can inform a policy. These connections motivate a conceptual framework without prescribing a fixed sequence of separate models.

A.3 Extended Roadmap Taxonomy

The roadmap in the main paper compresses a large amount of literature into a small number of stages. Table 5 expands this roadmap by identifying the dominant representation at each stage, its role in Physical AI, and representative works. The table is intended to make explicit the hidden continuity between fields that are often discussed separately: NLP world knowledge, multimodal representation learning, robot action modeling, model-based prediction, policy learning, and embodied deployment.

A key design choice in this taxonomy is to organize methods by the interface through which knowledge becomes physically useful. LLM-based world knowledge is primarily textual and parametric. Multimodal

Roadmap Component	Focus of This Analysis	Less Detailed Coverage
LLM-based world knowledge	Semantic, commonsense, procedural, causal, spatial, and affordance priors encoded in LLMs (Petroni et al., 2019; Roberts et al., 2020; Kandpal et al., 2023; Chang et al., 2024; Ruis et al., 2025)	General factual recall, knowledge editing, or memory analysis unrelated to physical reasoning
Multimodal grounding	VLMs/MLLMs that ground language-derived knowledge into images, videos, regions, objects, spatial relations, and affordances (Radford et al., 2021; Alayrac et al., 2022; Li et al., 2023b; Liu et al., 2023c; Team et al., 2023)	Generic captioning, VQA, or multimodal dialogue not tied to physical grounding or interaction
Action grounding	VLA models, action representations, policy learning, and language-conditioned embodied control (Driess et al., 2023; Zitkovich et al., 2023; Kim et al., 2025; Black et al., 2025; Physical Intelligence et al., 2025b)	Classical robot control, motion planning, or manipulation methods without foundation-model grounding
World models	Video, latent, interactive, and action-conditioned models that support prediction, simulation, planning, or policy learning (Ha and Schmidhuber, 2018; Hafner et al., 2019, 2020; Schrittwieser et al., 2020; Agarwal et al., 2025)	All video generation, all simulators, or all model-based RL methods outside the Physical AI roadmap
Embodied systems	Systems that close the loop between perception, planning, execution, recovery, and evaluation (Team et al., 2025b,a; NVIDIA et al., 2025; Intelligence et al., 2026)	Hardware-specific robot engineering, robot design, and domain-specific control stacks

Table 3 Scope of the conceptual framework. Categories describe the selected literature’s relation to language priors, grounded behavior, and prediction; neighboring topics remain relevant to Physical AI and Physical AGI.

Organizing Question	Focus	Connection to This Framework
How does AI connect to physical systems? (Li et al., 2023c; Dewi et al., 2025)	Concepts, applications, industrial systems, and cyber-physical perspectives	Establishes the broad deployment setting and the role of sensing, infrastructure, and control
How can visual models represent and generate physical processes? (Liu et al., 2025a)	Physics-aware generation, visual simulation, and physically grounded computer vision	Connects perceptual evidence and generated futures to physical consistency
How do representations and instructions become robot actions? (Zhang et al., 2026c; Gao et al., 2026; Kim et al., 2025)	Robot policies, action representations, and embodied control	Locates the action interface at which task knowledge meets embodiment constraints
How can learned predictions support decisions? (Ha and Schmidhuber, 2018; Hafner et al., 2020; Schrittwieser et al., 2020; Agarwal et al., 2025)	Prediction, latent dynamics, model-based planning, simulation, and policy learning	Provides mechanisms for comparing possible consequences and adapting to observed changes
How does prior knowledge become reusable physical competence? (This work)	Conceptual connections among grounding, action, prediction, learning, and integrated agents	Proposes evaluation questions for novelty, transfer, retention, and autonomy under explicit budgets

Table 4 Connections among research perspectives. The rows describe organizing questions and complementary contributions; they do not rank coverage, capability, or scientific quality.

grounding introduces visual and spatial representations. Action grounding introduces action tokens, trajectories, chunks, skills, and continuous controls. World modeling introduces future states, latent dynamics, rewards, values, or action-conditioned transitions. Policy learning then converts these representations into executable behavior, and embodied deployment tests whether the full stack can operate under feedback, noise, and real-world constraints.

These categories describe functions, not mutually exclusive model families. Language and multimodal models can expose structured or dense representations, policies may encode dynamics implicitly, and joint architectures can learn action and prediction together. For Physical AGI, the empirical question is whether the resulting agent learns and transfers efficiently under novelty; its placement in the taxonomy does not answer that question.

The representative works in the last column are selected as anchors rather than exhaustive lists. Many systems occupy multiple stages: for instance, a VLA system may combine multimodal grounding, action tokenization, policy learning, and real-world evaluation. We place each representative work according to its most salient role in the roadmap, while acknowledging that frontier systems increasingly blur these boundaries.

A.4 Extended Taxonomy of World Models

World models are used differently across reinforcement learning, video generation, robotics, autonomous driving, and Physical AI. In model-based reinforcement learning, a world model often refers to a learned transition or reward model used for planning. In video generation, the term is increasingly used for models that generate plausible future frames or interactive visual environments. In robotics and embodied AI, a world model should support action-conditioned prediction, counterfactual reasoning, recovery, and policy learning. These meanings overlap but are not identical.

Table 6 organizes predictive approaches by target and decision interface. Some categories overlap, and representation-learning precursors appear alongside models used for temporal prediction. The distinction relevant to the framework is what a model predicts and how that prediction supports a decision, rather than whether it carries the world-model label.

Stage	Main Representation	Role in Physical AI	Representative Works
LLM-based world knowledge	Textual and parametric knowledge	Provides semantic, commonsense, procedural, causal, spatial, and affordance priors	LAMA, closed-book QA, factual recall, procedural knowledge (Petroni et al., 2019; Roberts et al., 2020; Kandpal et al., 2023; Ruis et al., 2025)
Multimodal grounding	Image/video-language representations	Grounds world knowledge into objects, scenes, spatial relations, temporal events, and affordances	CLIP, Flamingo, BLIP-2, LLaVA, Gemini (Radford et al., 2021; Alayrac et al., 2022; Li et al., 2023b; Liu et al., 2023c; Team et al., 2023)
Action grounding	Action tokens, trajectories, chunks, skills, or continuous controls	Maps perception and language instructions to executable actions	PaLM-E, RT-2, OpenVLA, π_0 , $\pi_{0.5}$ (Driess et al., 2023; Zitkovich et al., 2023; Kim et al., 2025; Black et al., 2025; Physical Intelligence et al., 2025b)
World modeling	Future pixels, latent states, rewards, values, or action-conditioned transitions	Predicts and simulates possible futures for planning, policy learning, and counterfactual reasoning	World Models, Dreamer, MuZero, Genie, Cosmos, V-JEPA (Ha and Schmidhuber, 2018; Hafner et al., 2020; Schrittwieser et al., 2020; Bruce et al., 2024; Agarwal et al., 2025; Bardes et al., 2024)
Policy learning	Learned policies, action experts, diffusion/flow policies, or controllers	Converts perception, reasoning, and prediction into behavior	ACT, FAST, RDT-1B, GR00T N1 (Zhao et al., 2023; Pertsch et al., 2025; Liu et al., 2025c; NVIDIA et al., 2025)
Embodied deployment	Closed-loop systems with sensing, planning, execution, verification, and recovery	Tests whether models can reliably act in real or interactive environments	Gemini Robotics, RoboCasa, LIBERO, EmbodiedBench (Team et al., 2025b,a; Nasiriany et al., 2024; Liu et al., 2023b; Yang et al., 2025)

Table 5 Extended taxonomy of the roadmap from LLM-based world knowledge to Physical AI.

Category	Prediction Target	Role in Physical AI	Representative Works
Classical / model-based RL world models	Future states, rewards, values, or policy-relevant quantities	Planning, latent imagination, decision making, and policy improvement	World Models, PlaNet, Dreamer, DreamerV3, MuZero (Ha and Schmidhuber, 2018; Hafner et al., 2019, 2020, 2023; Schrittwieser et al., 2020)
Video-space world models	Future pixels, frames, or video tokens	Visual imagination, future scene prediction, synthetic data, and simulated experience	GAIA-1, UniSim, Genie, Cosmos (Hu et al., 2023; Yang et al., 2024; Bruce et al., 2024; Agarwal et al., 2025)
Latent / representation- space world models	Latent states, embeddings, or masked spatial and temporal representations	Efficient long-horizon prediction, compact planning, and control-relevant representation learning	PlaNet, Dreamer, I-JEPA, V-JEPA, V-JEPA 2 (Hafner et al., 2019, 2020; Assran et al., 2023; Bardes et al., 2024; Assran et al., 2025)
Interactive / action-conditioned world models	Future observations or latent states conditioned on candidate actions	Counterfactual reasoning, simulation-based policy learning, safety evaluation, and recovery	MuZero, UniSim, Genie, GAIA-1, V-JEPA 2, Cosmos (Schrittwieser et al., 2020; Yang et al., 2024; Bruce et al., 2024; Hu et al., 2023; Assran et al., 2025; Agarwal et al., 2025)
World foundation models for Physical AI	General-purpose predictive or generative world representations	Adaptable substrate for robotics, autonomous driving, embodied agents, and synthetic data	Cosmos, Genie-style models, V-JEPA-style models (Agarwal et al., 2025; Bruce et al., 2024; Bardes et al., 2024; Assran et al., 2025)

Table 6 Extended taxonomy of world models for Physical AI. The categories are organized by prediction target and their function in the roadmap from LLM-based world knowledge to embodied action.

A.4.1 Classical and decision-centric world models.

Classical world models are rooted in model-based reinforcement learning and planning. They learn transition, reward, value, or policy-relevant predictions that allow agents to plan before acting. World Models, PlaNet, Dreamer, DreamerV3, and MuZero establish this foundation by showing that agents can learn compact internal models and use them for imagination, planning, and policy improvement (Ha and Schmidhuber, 2018; Hafner et al., 2019, 2020, 2023; Schrittwieser et al., 2020). For Physical AI, this line provides the decision-making substrate: agents should not only react to observations, but also evaluate possible futures.

A.4.2 Video-space world models.

Video-space world models predict future visual observations. They are attractive for Physical AI because videos expose motion, temporal evolution, scene changes, and possible future outcomes. GAIA-1 models autonomous-driving futures from video, text, and action inputs (Hu et al., 2023); UniSim learns an interactive real-world simulator from heterogeneous data and uses it for policy training (Yang et al., 2024); Genie learns generative interactive environments from unlabelled videos (Bruce et al., 2024); and Cosmos positions world foundation models as adaptable world models for Physical AI (Agarwal et al., 2025). The main limitation is that visual realism does not guarantee physical correctness. A generated rollout may look plausible while violating object permanence, contact constraints, controllability, or causal consistency.

A.4.3 Latent and representation-space world models.

Latent world models predict in compact representation spaces rather than pixel space. They can reduce the cost of prediction by avoiding reconstruction of every visual detail, provided the learned representation preserves what the decision requires. JEPA-style work develops representation-space prediction, with I-JEPA providing a spatial representation-learning precursor and V-JEPA extending prediction to video (LeCun, 2022; Assran et al., 2023; Bardes et al., 2024). V-JEPA 2 extends this line to video-scale learning and post-trains an action-conditioned latent world model for robot planning (Assran et al., 2025). Planning speed, rollout accuracy, and uncertainty estimates still require evaluation in the intended downstream setting.

A.4.4 Interactive and action-conditioned world models.

When a predictive model is used to compare candidate actions, its output must depend appropriately on those actions and retain the consequences relevant to the task. This requires testing action sensitivity, temporal consistency, and physical plausibility in addition to appearance. Forecasts can inform planning, policy learning, and recovery, but their benefit depends on model error and the agent’s response to uncertainty. Passive predictive models can also support representation learning or state estimation without themselves providing an action-conditioned simulator.

A.4.5 Relation to LLM-based world knowledge.

Language priors and predictive mechanisms can play complementary roles: the former suggest goals, procedures, and plausible events, while the latter relate a specified state and possible intervention to measurable consequences. A shared architecture may serve both functions. The connection matters for Physical AGI when a prediction or observation changes how the agent solves an unfamiliar task, or when the resulting experience improves later learning and transfer.

A.5 Benchmarks and Evaluation Protocols

Evaluation is a central difficulty for Physical AI because the roadmap spans several different kinds of competence. Static language benchmarks can test whether a model encodes commonsense or procedural knowledge, but they cannot determine whether that knowledge is grounded in a physical state. Vision-language benchmarks can test perception and grounding, but they often stop at recognition or description. VLA benchmarks can test whether actions are predicted from observations and instructions, but open-loop action accuracy does not fully capture closed-loop execution. World-model benchmarks can test prediction, but prediction quality must ultimately be judged by whether it supports planning and control.

Table 7 relates representative evaluations to the roadmap’s interfaces. Sections 2 and 8 add the generality criteria and benchmark-coverage analysis: closed-loop completion is necessary evidence about execution, while progress towards Physical AGI additionally requires held-out novelty, learning efficiency, transfer, and retention under disclosed budgets.

Roadmap Stage	Benchmark / Evaluation Type	What to Evaluate	Representative Works
LLM world knowledge	Physical commonsense, tool understanding, factual/procedural knowledge	Whether LLMs encode usable semantic, commonsense, procedural, and causal priors	PHYBench, PhySense, PhysToolBench (Qiu et al., 2025; Xu et al., 2025; Zhang et al., 2025c)
VLM/MLLM grounding	Spatial, temporal, affordance, and physical reasoning benchmarks	Whether language-derived knowledge is grounded into perception, spatial relations, and physical states	BLINK, Video-MME, PhysBench, QuantiPhy, MASS-Bench (Fu et al., 2024, 2025; Chow et al., 2025; Puyin et al., 2025; Wu et al., 2025)
VLA / action grounding	Robot manipulation, navigation, and action-prediction benchmarks	Whether models can map instructions and observations to executable actions	RT-2, OpenVLA, LIBERO, LIBERO-Pro (Zitkovich et al., 2023; Kim et al., 2025; Liu et al., 2023b; Zhou et al., 2025)
World models	Video prediction, latent prediction, action-conditioned simulation, planning evaluation	Whether models predict physically plausible, controllable, temporally consistent futures	World Models, Dreamer, Genie, Cosmos, V-JEPA 2 (Ha and Schmidhuber, 2018; Hafner et al., 2020; Bruce et al., 2024; Agarwal et al., 2025; Assran et al., 2025)
Embodied agents	Closed-loop simulated or real-world tasks	Whether systems complete tasks, recover from errors, and generalize across environments and embodiments	BEHAVIOR, EAI, EmbodiedBench, RoboSuite, RoboCasa (Srivastava et al.; Li et al., 2024; Yang et al., 2025; Zhu et al., 2020; Nasiriany et al., 2024)
Closed frontier systems	Black-box or product-level evaluation	Capability, reliability, reproducibility, safety, and transparency under limited disclosure	Gemini Robotics, Gemini Robotics 1.5, GR00T N1, π -series systems (Team et al., 2025b,a; NVIDIA et al., 2025; Black et al., 2025; Physical Intelligence et al., 2025b; Intelligence et al., 2026)

Table 7 Evaluation protocols along the roadmap. Evaluation of embodied generality should extend static recognition and offline prediction with closed-loop task completion, robustness, safety, recovery, and cross-embodiment generalization.

A useful evaluation suite should therefore include both stage-specific and system-level metrics. Stage-specific metrics diagnose where a system fails: factual or physical commonsense, perceptual grounding, action prediction, world-model rollout, or closed-loop execution. System-level metrics evaluate whether these components work together under deployment constraints. For example, a strong VLA may still fail if its actions accumulate error, if its world model produces visually plausible but physically inconsistent futures, or if its controller cannot recover from perturbations. This is why task success, intervention count, robustness, safety, and recovery should be reported alongside conventional accuracy or prediction metrics.

A.6 Frontier Systems and Closed Models

Many influential Physical AI systems are released as frontier products, platforms, or partially documented technical reports rather than fully open academic artifacts. This creates a gap between real-world usage and academic evaluation. Closed or partially disclosed systems may demonstrate important capabilities, shape the terminology of the field, and influence user expectations, but their training data, architecture details, evaluation protocols, and failure cases are often unavailable.

Table 8 records representative systems through the cited reports and documentation. Their roles motivate black-box evaluation and careful attribution when training data, internal components, or human assistance are incompletely disclosed. The categories refer to the cited releases, since access conditions and available artifacts can change; inclusion does not establish general physical competence.

System	Category	Openness / Citation Type	Role in the Roadmap
GPT-4 (Achiam et al., 2023)	Multimodal language model	Closed / technical report	Language and multimodal capabilities relevant to prior knowledge and reasoning; the report does not establish integrated robot competence
Claude-style assistants (Anthropic, 2024)	LLM / agentic assistant	Closed / product documentation	Reasoning, tool use, coding, and agentic orchestration
Gemini Robotics and Gemini Robotics 1.5 (Team et al., 2025b,a)	Robotics foundation model	Closed or partially disclosed / technical report	Multimodal reasoning, embodied control, and real-world robot interaction
Cosmos (Agarwal et al., 2025)	World foundation model platform	Partially open / technical report	World modeling, synthetic data, simulation, autonomous driving, and robotics
GR00T N1 (NVIDIA et al., 2025)	Humanoid foundation model	Partially open / technical report	Generalist humanoid policies and cross-embodiment action learning
π -series systems (Black et al., 2025; Physical Intelligence et al., 2025b; Intelligence et al., 2026)	Generalist VLA / robot foundation models	Partially disclosed / technical reports	Action grounding, open-world generalization, policy learning, and embodied deployment

Table 8 Representative systems and their disclosure in the cited releases. These examples motivate product-level and black-box evaluation protocols; openness and access can differ across versions.

The main challenge posed by closed systems is not only that they are difficult to reproduce. It is also that they may combine several roadmap stages into a single product-level stack, making ablation and attribution difficult. A system may appear to have strong physical reasoning because of its LLM prior, its perception module, its action policy, its retrieval system, its simulator, or its human-feedback pipeline. Without transparent interfaces and standardized black-box tests, it is difficult to identify which component contributes to success or failure. This motivates evaluation protocols that separate capability testing, robustness testing, safety testing, and reproducibility reporting.

A.7 Failure Modes Along the Roadmap

The roadmap is useful not only because it organizes progress, but also because it localizes failures. A Physical AI system may fail at the level of world knowledge, perception, action, prediction, policy learning, deployment, or evaluation. These failures are qualitatively different. An LLM hallucination produces a plausible but ungrounded plan; a VLM grounding failure misidentifies the state of the world; a VLA failure maps a correct

goal to the wrong action; a world-model failure predicts an implausible future; and a deployment failure can arise from sensing, latency, calibration, or controller mismatch.

Table 9 summarizes representative failure modes. These failures motivate our deployment-oriented discussion of challenges in the main paper. The table also clarifies why Physical AI cannot be evaluated by a single benchmark: each roadmap stage requires different diagnostics, and end-to-end task success alone may hide the source of failure.

Component	Typical Failure Mode	Why It Matters for Physical AI
LLMs	Hallucinated or ungrounded physical knowledge; overconfident plans; missing metric state	The model may propose plausible language plans that violate geometry, contact, force, or object-state constraints
VLMs / MLLMs	Correct semantic description but weak dense grounding	The model may identify objects but fail to estimate pose, depth, uncertainty, reachability, or action-conditioned dynamics
VLAs	Poor cross-embodiment generalization; data-dependent policies; brittle recovery	The same instruction may require different grasps, trajectories, or control strategies across robots and environments
World models	Visually plausible but physically inconsistent futures	Photorealistic generation may still violate object permanence, contact, gravity, controllability, or causal dynamics
Policy learning	Offline success but closed-loop failure	A model may predict correct actions under dataset states but fail under compounding errors or real-time perturbations
Embodied systems	Sensor, calibration, latency, controller, or hardware failures	Physical performance depends on the full system stack, not only model accuracy
Closed frontier systems	Limited reproducibility and incomplete disclosure	Strong product-level systems can shape the field while being difficult to benchmark, ablate, or compare fairly

Table 9 Representative failure modes along the roadmap from LLM-based world knowledge to Physical AI.

The failure-mode view also suggests a practical debugging strategy. If a system fails before action, the issue may lie in world knowledge or perceptual grounding. If it fails during action, the issue may lie in action representation, embodiment transfer, or controller design. If it fails after several steps, the issue may lie in world modeling, compounding error, memory, or recovery. If it succeeds in simulation but fails in the real world, the issue may lie in sim-to-real transfer, sensing, calibration, latency, or hidden deployment assumptions. This decomposition turns the roadmap into an evaluation tool rather than just a taxonomy.

A.8 Extended Discussion of Challenges and Future Directions

The main paper identifies six generality gaps. Here we relate them to technical interfaces in existing work: the information preserved by grounding, the transfer supported by action representations, the decision value of prediction, and the reliability of integrated behavior. These interfaces may be implemented by separate modules or within a joint model. Their evaluation must make task novelty, adaptation, retention, and human assistance explicit.

A.8.1 Implicit world knowledge and dense physical grounding.

LLM-based world knowledge is broad, but it is not an explicit symbolic database. It is stored as parametric regularities and exposed through prompting, context, plans, programs, or tool calls. Knowledge-probing and closed-book QA studies show that language models can store factual and relational knowledge in parameters (Petroni et al., 2019; Roberts et al., 2020), while later studies analyze long-tail factual acquisition, factual recall, and pretraining dynamics (Kandpal et al., 2023; Chang et al., 2024; Yuan et al., 2024). Materialization and mechanistic analyses further show that parts of such knowledge can be extracted or traced

through model computations (Hu et al., 2025; Geva et al., 2023; Tao et al., 2024). For physical decisions, the interface must preserve the quantities the task requires, which may include pose, reachability, contact, force, friction, uncertainty, and temporal dynamics. Procedural knowledge and task-level planning provide useful priors (Ruis et al., 2025; Huang et al., 2022; Ichter et al., 2023; Huang et al., 2023c; Liang et al., 2023; Singh et al., 2023; Song et al., 2023; Rana et al., 2023), but they remain insufficient without perceptual grounding and physical verification.

A.8.2 Multimodal grounding and physically faithful perception.

VLMs and MLLMs offer a bridge from language-mediated priors to perceptual observations. Contrastive and multimodal pretraining align images or videos with language (Radford et al., 2021; Alayrac et al., 2022; Li et al., 2023b; Liu et al., 2023c; Team et al., 2023), enabling models to connect objects, scenes, and instructions to visual inputs. However, Physical AI requires more than captioning or visual QA. Depending on the task, it may require spatial and temporal grounding, affordance estimation, quantitative reasoning, or dense state estimation. Recent benchmarks and evaluations expose gaps in low-level visual perception, long-video understanding, physical reasoning, quantitative physics, and motion-aware spatiotemporal grounding (Fu et al., 2024, 2025; Chow et al., 2025; Puyin et al., 2025; Wu et al., 2025). Related work on dense physical perception and intermediate-feature grounding suggests that VLM representations may be reused for downstream Physical AI tasks, but their outputs must be transformed into action-relevant or prediction-relevant representations (Zhang et al., 2025a,b, 2026c,a).

A.8.3 VLA generalization and action-interface bottlenecks.

VLA models connect visual observations and language instructions to action outputs, making them a key interface between multimodal reasoning and embodied control. Representative systems such as PaLM-E, RT-2, OpenVLA, and the π -series demonstrate the promise of transferring web-scale or foundation-model knowledge into robotic action (Driess et al., 2023; Zitkovich et al., 2023; Kim et al., 2025; Black et al., 2025; Physical Intelligence et al., 2025b). Nevertheless, VLA policies face three persistent bottlenecks. First, action spaces vary across embodiments, including action tokens, end-effector poses, trajectories, action chunks, and continuous controls. Second, robot data remain much smaller and more heterogeneous than language or vision data. Third, imitation-trained policies can be brittle under distribution shift and may lack recovery behavior. Recent work on VLA learning, action abstraction, and generalist robot policies explores scalable action representations, data mixtures, and richer policy architectures (Gao et al., 2026; Zhang et al., 2026c; Zhao et al., 2023; Pertsch et al., 2025; Liu et al., 2025c; NVIDIA et al., 2025; Intelligence et al., 2026). A promising direction is to augment VLA policies with memory, LLM agents, or world models so that policies do not only map observations to actions, but also reason over goals, histories, and possible futures.

A.8.4 World models beyond video generation.

World models expose predictions or simulated consequences that can complement action policies; policies can also encode predictive structure implicitly or share a model with a prediction objective. Classical and decision-centric world models learn transition, reward, value, or policy-relevant quantities for planning and policy improvement (Ha and Schmidhuber, 2018; Hafner et al., 2019, 2020, 2023; Schrittwieser et al., 2020). Video-space world models generate or predict future visual observations and can support visual imagination, synthetic data, and interactive simulation (Hu et al., 2023; Yang et al., 2024; Bruce et al., 2024; Agarwal et al., 2025). Latent and JEPA-style world models instead predict in representation space, trading pixel-level reconstruction for efficiency and planning-relevant abstraction (LeCun, 2022; Assran et al., 2023; Bardes et al., 2024; Assran et al., 2025). This distinction is important: photorealistic generation does not guarantee physical correctness, while latent prediction may be efficient but difficult to interpret or use directly without task heads, decoders, or policy interfaces. For Physical AI, a useful world model should be temporally consistent, action-conditioned, controllable, physically plausible, and useful for planning or closed-loop decision making. Recent work on physically grounded world-model evaluation and language-guided latent prediction provides early steps toward this goal (Lin et al., 2026; Zhang et al., 2026a).

A.8.5 Deployment and system-level evaluation.

Even strong models can fail when deployed as Physical AI systems. World models and simulators can support training and planning, but real-world deployment introduces sensor noise, latency, calibration errors, embodiment mismatch, sim-to-real transfer, safety constraints, and recovery requirements. Robotics and embodied benchmarks such as LIBERO, RoboCasa, BEHAVIOR, RoboSuite, and EmbodiedBench evaluate different parts of this system-level challenge (Liu et al., 2023b; Nasiriany et al., 2024; Srivastava et al.; Zhu et al., 2020; Yang et al., 2025). Frontier systems such as Gemini Robotics, GR00T N1, and π -series models further show that Physical AI is becoming a product-level systems category, but many such systems are closed or only partially disclosed (Team et al., 2025b,a; NVIDIA et al., 2025; Black et al., 2025; Physical Intelligence et al., 2025b; Intelligence et al., 2026). Future evaluation should therefore report closed-loop task completion, recovery, intervention counts, robustness, safety, reproducibility, and cross-embodiment generalization rather than relying only on static model-level metrics.

Challenge	Interface Mismatch	Future Direction	Representative Evidence
Implicit world knowledge	LLM priors are language-mediated and difficult to convert into metric physical state	Extract, align, and ground semantic, procedural, and causal priors into dense physical representations	Knowledge probing, factual recall, procedural knowledge, language planning (Petroni et al., 2019; Roberts et al., 2020; Kandpal et al., 2023; Ruis et al., 2025; Ichter et al., 2023)
Physical perception	Sparse semantic interfaces can omit task-relevant state and uncertainty	Move toward spatial, temporal, affordance, quantitative, and action-relevant grounding	VLM/MLLM grounding and physical reasoning benchmarks (Radford et al., 2021; Liu et al., 2023c; Fu et al., 2024, 2025; Chow et al., 2025; Puyin et al., 2025)
Generalist VLA policies	Actions are embodiment-specific and robot data are limited	Develop scalable action representations, cross-embodiment transfer, and policies augmented with memory or world models	PaLM-E, RT-2, OpenVLA, π -series, FAST, GR00T N1 (Driess et al., 2023; Zitkovich et al., 2023; Kim et al., 2025; Black et al., 2025; Pertsch et al., 2025; NVIDIA et al., 2025)
Predictive world modeling	Video realism does not imply physical correctness; latent models may require task heads or policy interfaces	Build action-conditioned, controllable, efficient, and physically plausible world models	Dreamer, MuZero, Genie, UniSim, Cosmos, V-JEPA (Hafner et al., 2020; Schrittwieser et al., 2020; Bruce et al., 2024; Yang et al., 2024; Agarwal et al., 2025; Bardes et al., 2024)
Deployment	Model-level accuracy does not capture sensing, control, latency, recovery, safety, or sim-to-real robustness	Evaluate integrated systems with closed-loop task completion, recovery, safety, and reproducibility	LIBERO, RoboCasa, EmbodiedBench, Gemini Robotics, GR00T N1 (Liu et al., 2023b; Nasiriany et al., 2024; Yang et al., 2025; Team et al., 2025b; NVIDIA et al., 2025)

Table 10 Technical interfaces underlying the generality gaps in the main paper. The cited work provides examples relevant to each interface; the listed directions are prospective research questions.

A.9 Terminology

Term	Working Definition
World knowledge	Semantic, commonsense, procedural, causal, spatial, and affordance priors about objects, agents, actions, environments, and likely consequences.
LLM-based world knowledge	Language-mediated world priors stored as parametric regularities in LLMs and exposed through prompting, context, or agentic reasoning.
World model	A predictive or simulative model that estimates future observations, latent states, rewards, values, or action consequences from current states and possible actions.
VLA model	A model that maps visual observations, language instructions, and sometimes embodiment states into executable actions or action-relevant representations.
Physical AI	AI systems for perception, prediction, reasoning, or action in physical environments, including embodied agents and models used to study physical interaction. No particular model family is required.
Physical AGI	An aspirational regime of broad cognitive and physical competence in which an agent efficiently acquires, composes, transfers, and retains skills across unfamiliar tasks, environments, dynamics, and embodiments under an explicit task-specific experience budget.
Adaptation budget	The task-specific instruction, demonstrations, interaction, computation, and human assistance available when learning or adapting to a held-out condition.
Generality evidence	Measured breadth, learning efficiency, transfer, and retention under disclosed conditions; an architectural label or component list does not certify Physical AGI.

Table 11 Working terminology. Physical AGI denotes a research aspiration; the definitions specify evaluation questions and do not establish a validated certification threshold.

Table 11 records functional definitions for concepts whose meanings vary across communities. In particular, world knowledge describes prior information, while a world model describes a predictive function; one implementation can provide both. Physical AI identifies the broad domain, whereas Physical AGI names an aspirational capability regime whose assessment depends on learning and transfer evidence under explicit budgets.